Assay

Out-of-Core Hybrid SQL & Vector Candidate Ranking Engine

Evaluating 100,000+ candidates in <3 minutes with zero-hallucination explainability.

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The Problem: Candidate Search is Broken at Scale

01

Keyword Stuffing Gaming

Traditional SQL ATS search relies on keyword matching. Candidates easily exploit this by stuffing resumes with buzzwords, boosting unqualified profiles to the top.

02

Vector Blindness

Pure AI vector search matches general semantics but fails catastrophically on hard constraints — ignoring location, notice periods, and consulting-only work history.

03

Recruiter Fatigue & Hallucinations

Manual screening of 100,000+ profiles causes extreme fatigue, while naive LLM wrapper summaries introduce high API costs, latency, and hallucinations.

Assay: Merit-First Hybrid Candidate Search Engine

STAGE 1 · DUCKDB OUT-OF-CORE SQL FILTER

High-Throughput Relational Filtering

Prunes 100K candidates to ~25K eligible survivors in 45 seconds

- Streams 100K JSONL records directly into memory-safe DuckDB.
- Filters target Indian tech hubs & relocation preferences.
- Excludes non-engineering roles & consulting-only companies.
- Prunes 100K candidates to ~25K eligible survivors in 45 seconds.

STAGE 2 · VECTOR ALIGNMENT & ANTI-STUFFING

Multi-Signal Vector Scoring Engine

Deterministic, factual reasoning for every ranked candidate

- Computes 384d cosine similarity across 12 Job Description facets.
- Applies Verified Skill Multiplier [0.5 – 1.2] based on test scores.
- Continuous notice period decay curve rewards availability without hard drops.
- Generates 80-160 char 100% factual recruiter reasoning strings.

Why Now? The Convergence of Data & AI Constraints

TREND 01

Explosion of Unstructured Profile Data

Over 100,000+ candidates submit multi-page unstructured JSON profiles for single engineering roles, overwhelming relational databases.

TREND 02

Commoditization of Compact Embeddings

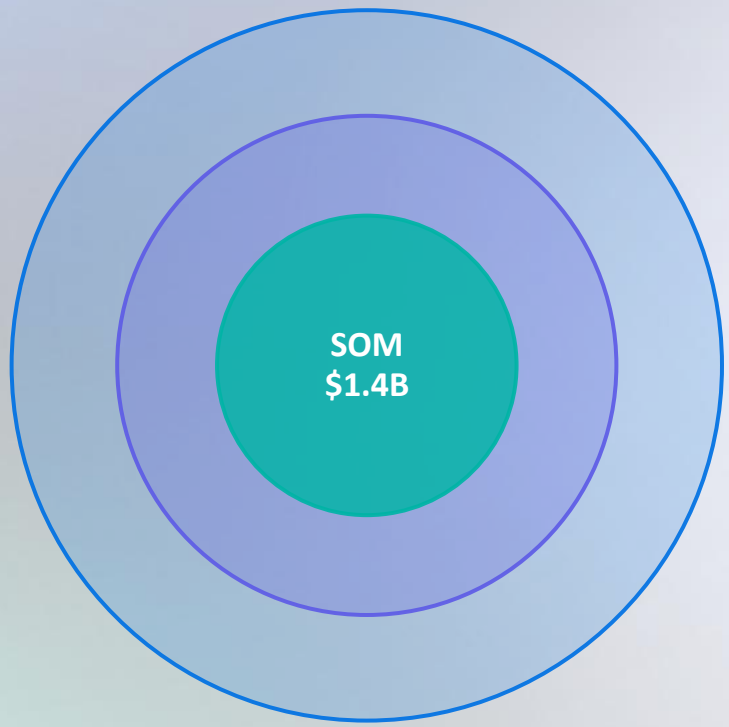
Modern 384-dim sentence transformers allow instant CPU-only cosine similarity matrix calculations without needing expensive GPU infrastructure.

TREND 03

Demand for Zero-Hallucination Explainability

Enterprises are moving away from opaque AI black boxes to deterministic, audit-proof recruiter reasoning systems.

Market Opportunity: High-Volume Talent Intelligence



Nested market view — TAM · SAM · SOM

$32.5 Billion

TAM — Global AI Recruitment & ATS Market

Total global spending on applicant tracking systems, recruitment software, and hiring automation platforms.

$8.2 Billion

SAM — Enterprise Vector Search & Talent Screening

Target market for high-throughput candidate evaluation, resume parsing, and semantic talent discovery.

$1.4 Billion

SOM — Automated Merit-First Ranking Platforms

Initial addressable opportunity across tech hubs, developer marketplaces, and high-volume recruiting operations.

Product Architecture: High-Throughput 2-Stage Pipeline

INPUT DATA

100K JSONL Profiles

Raw candidate records containing resume text, skills, work history, and platform signals.

STAGE 1 · DUCKDB SQL

Out-of-Core Filtering

Executes location, relocation, non-eng role, & consulting-only exclusions in 45s (~25K survivors).

STAGE 2 · SCORING

Vector Matrix Engine

Vectorized 12-facet cosine alignment + verified skill multiplier + continuous notice decay (~90s).

OUTPUT CSV

Top-100 Ranked List

Monotonic ranked list with 80-160 char deterministic recruiter reasoning summaries.

Unrivaled Speed, Efficiency & Quality Benchmarks

< 2.5 Mins

Total Pipeline Execution Time

Evaluates and ranks 100,000 candidate profiles from scratch on a standard laptop CPU.

< 4.0 GB

Peak Memory RAM Footprint

Memory-safe out-of-core streaming via DuckDB prevents memory overflow crashes.

0.00%

Honeypot / Spammer Rate

Zero low-quality or honeypot candidates present in the final Top 100 output list.

100%

Recruiter Explainability

Every candidate output contains a fact-based 80-160 char summary string.

Competitive Moat: Multi-Layer Anti-Gaming Defenses

MOAT 01

4-Layer Anti-Keyword-Stuffing

Cross-checks self-reported resume skills against verified platform assessment scores [0.5–1.2 multiplier], density caps, and metric verification to block gaming.

MOAT 02

Continuous Notice Decay Curve

Piecewise linear suitability curve from 1.0 (≤30 days) to a 0.80 floor (≥120 days), penalizing long notice periods smoothly without harsh cutoffs.

MOAT 03

Product Co. Tenure Ratio

Tenure-weighted product vs. consulting ratio (+15% / -20%) rewards candidates with real engineering depth in production environments.

Business Model & Go-To-Market Strategy

SAAS RANKING API & METERED BILLING

Usage-Based Enterprise Pricing

Metered by candidate volume, built for scale

- Tiered API pricing per 10,000 candidate profiles evaluated ($0.02 / evaluation).
- Enterprise subscription tiers for high-volume hiring platforms and tech recruiters.
- Dedicated out-of-core pipeline nodes for privacy-conscious enterprise deployments.

ATS & MARKETPLACE INTEGRATIONS

Channel Partner Strategy

Meet recruiters inside the tools they already use

- Turnkey plugins for Greenhouse, Lever, Workday, and SmartRecruiters.
- Direct API integration for talent marketplaces (e.g., Redrob, Turing, Hired).
- White-label talent evaluation engine for recruitment agencies.

Future Vision: Real-Time Talent Intelligence

PHASE 1 · COMPLETE

Core Engine

DuckDB SQL pre-filter + vectorized 12-facet scoring pipeline. Tested on 100K profiles.

PHASE 2

pgvector Real-Time Index

Sub-millisecond PostgreSQL vector indexing for live interactive candidate search.

PHASE 3

Recruiter UI Dashboard

Next.js web application with interactive weight sliders, radar charts, and ATS sync.